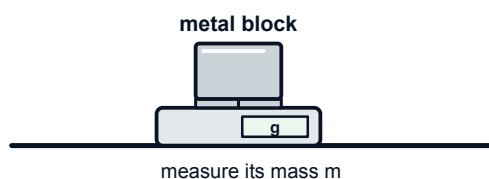


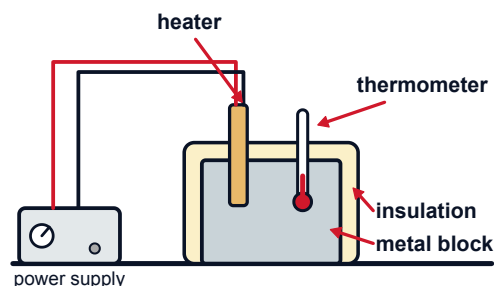
Specific heat capacity

A student measures the specific heat capacity of a metal block.

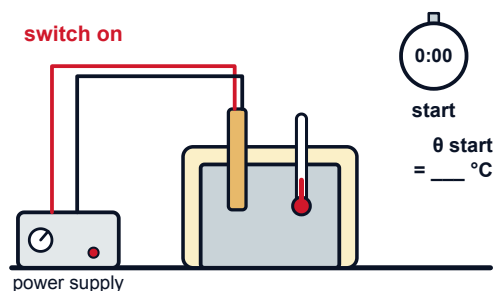
- 1** Measure the mass of the block with a balance.



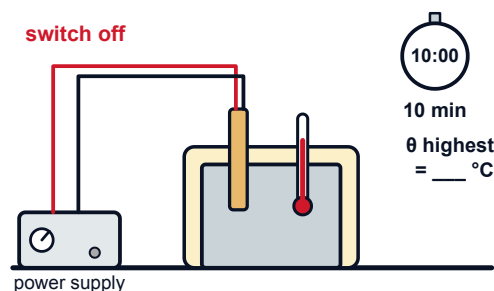
- 2** Put a heater and a thermometer into the block and wrap it in insulation.



- 3** Record the starting temperature, then switch on the heater and start a timer.



- 4** After 10 minutes, switch off and record the highest temperature reached.



- 5** Calculate the energy supplied from the power of the heater and the time.

$$E = P \times t$$

energy (J) = power (W) × time (s)

time in seconds:

10 minutes = 600 s

- 6** Calculate the specific heat capacity: energy ÷ (mass × temperature rise).

$$c = E \div (m \times \Delta\theta)$$

J/kg°C = J ÷ (kg × °C)

$$\Delta\theta = \theta_{\text{highest}} - \theta_{\text{start}}$$

m in kg (not g)